

Processing Peaker Plant Operations Data in Pittsfield, MA

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2026-02-02

R Markdown

Hourly emissions data for the Pittsfield peaker power plant (ID: 50002) was downloaded from the EPA CAMPD website. Timeline: April 1, 2024 to September 30, 2025. The goal of this script is to clean the raw hourly emissions data for analysis.

```
#provide the file path for the original peaker hourly emissions data from April 1, 2024 to September 30
peakeropcsv <- fread("data/raw/peaker-hourly-emissions-240401-250930.csv")
```

```
#rename columns to be more code-friendly
peakerop <- peakeropcsv |> dplyr::rename(state = State,
                                         peakername = "Facility Name",
                                         peakerid = "Facility ID",
                                         date = Date,
                                         hour = Hour,
                                         grossload_mw = `Gross Load (MW)`,
                                         steamload_1000lbsperhr = `Steam Load (1000 lb/hr)`,
                                         so2mass_lbs = `SO2 Mass (lbs)`,
                                         co2mass_shorttons = `CO2 Mass (short tons)`,
                                         noxmass_lbs = `NOx Mass (lbs)`,
                                         heatinput_mmbtu = `Heat Input (mmBtu)`)
```

```
peakerop$operating <- ifelse(!is.na(peakerop$heatinput_mmbtu), "1", "0")
#1 indicates yes that the peaker was operating, 0 indicates no
```

```
table(peakerop$operating)
```

```
##
##      0      1
## 11647 1505
```

```
print("The peaker ran for 1505 hours between April 1, 2024 and September 30, 2025.")
```

```
## [1] "The peaker ran for 1505 hours between April 1, 2024 and September 30, 2025."
```

```
class(peakerop$date)
```

```
## [1] "IDate" "Date"
```

```
#extract the year, month, and day columns using lubridate
peakerop$year <- year(peakerop$date)
peakerop$month <- month(peakerop$date)
peakerop$day <- day(peakerop$date)
```

```
#create a unique join column in the format YYYY;MM;DD;HH
peakerop <- peakerop %>%
```

```

mutate(
  datetime = paste0(
    year(date), ";",
    sprintf("%02d", month(date)), ";",
    sprintf("%02d", day(date)), ";",
    sprintf("%02d", hour)
  )
)

head(peakerop$datetime)

## [1] "2024;04;01;00" "2024;04;01;01" "2024;04;01;02" "2024;04;01;03"
## [5] "2024;04;01;04" "2024;04;01;05"

class(peakerop$month)

## [1] "numeric"

#create a season column based on the month column
peakerop$season <- ifelse(peakerop$month %in% c(12, 1, 2), "Winter",
  ifelse(peakerop$month %in% c(3, 4, 5), "Spring",
    ifelse(peakerop$month %in% c(6, 7, 8), "Summer",
      ifelse(peakerop$month %in% c(9, 10, 11), "Fall",
        NA))))))

peakerop |> dplyr::filter(season == "Summer") |> count(operating)

##      operating      n
##      <char> <int>
## 1:         0  3021
## 2:         1  1395

peakeropon <- peakerop |> dplyr::filter(operating == "1")

table(peakeropon$month)

##
##  1  2  5  6  7  8  9 10 12
## 19 13 22 289 744 362 26 23  7

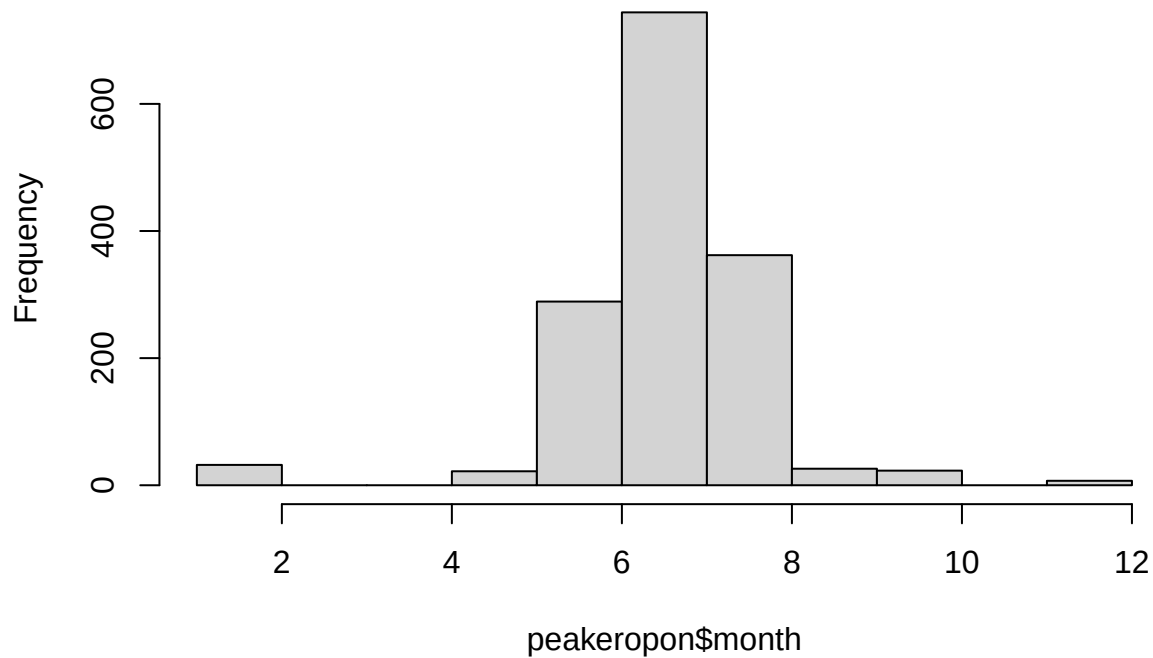
table(peakeropon$season)

##
##  Fall Spring Summer Winter
##    49    22   1395    39

hist(peakeropon$month)

```

Histogram of peakeropon\$month



#still need to add weekend column and time of day column

```
season_counts <- peakerop %>%
  count(season, operating)

ggpeakerseasonly <- ggplot(season_counts, aes(x = season, y = n, fill = factor(operating))) +
  geom_col(position = "dodge") +
  labs(
    title = "Peaker plant hourly operations by season",
    x = "Season",
    y = "Count",
    fill = "Operating\n(1 = Yes, 0 = No)"
  ) +
  theme_minimal()

ggsave("plots/ggpeakerseasonly.png", ggpeakerseasonly, height = 6, width = 8, units = "in")

peakerop1 <- peakerop %>%
  mutate(
    season = factor(season, levels = c("Winter", "Spring", "Summer", "Fall"))
  )

peakerop1 <- peakerop1 %>%
  mutate(
    season_year = paste(season, year)
  )
```

```

peakerop1 <- peakerop1 %>%
  arrange(year, season) %>% # ensures chronological ordering
  mutate(
    season_year = factor(season_year, levels = unique(season_year))
  )

season_year_counts <- peakerop1 %>%
  count(season_year, operating)

ggpeakerseason <- ggplot(season_year_counts,
  aes(x = season_year, y = n, fill = factor(operating))) +
  geom_col(position = "dodge") +
  labs(
    title = "Number of hours that the peaker plant was operating over time",
    x = "Season-Year",
    y = "COunt",
    fill = "Operating\n(1 = Yes, 0 = No)"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )

ggsave("plots/ggpeakerseason.png", ggpeakerseason, width = 8, height = 6, units = "in")

#NOTE TO SELF: WINTER 2024 and WINTER 2025 should be chronologically near each other.

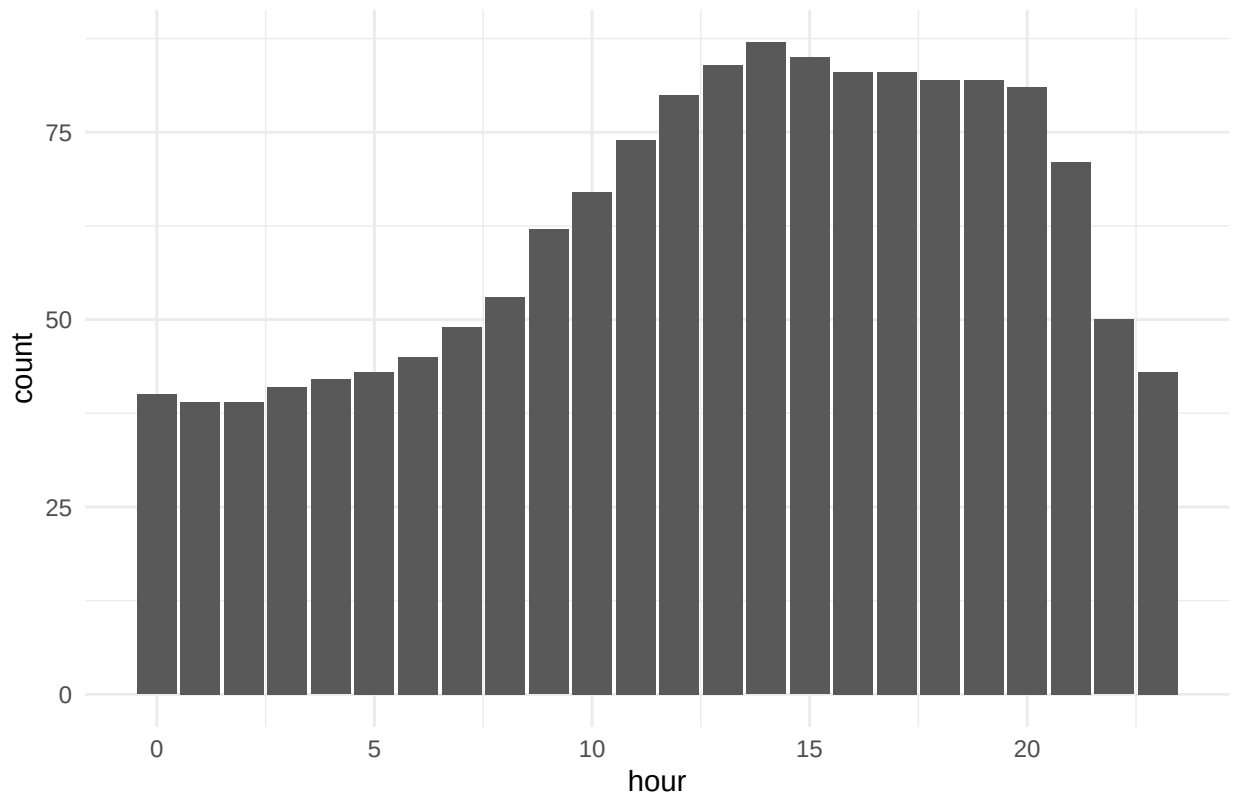
#create a time of day column based on hour column
#what hours is the peaker on?
table(peakeropon$hour)

##
## 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
## 40 39 39 41 42 43 45 49 53 62 67 74 80 84 87 85 83 83 82 82 81 71 50 43

ggplot(peakeropon, aes(x = hour)) +
  geom_bar() +
  labs(title = "How many times is the peaker on at each hour?") +
  theme_minimal()

```

How many times is the peaker on at each hour?



```
#create time of day column based on hour column
peakerop$timeofday <- ifelse(peakerop$hour %in% c(0, 1, 2, 3), "Late Night",
                             ifelse(peakerop$hour %in% c(4, 5, 6, 7), "Early Morning",
                                     ifelse(peakerop$hour %in% c(8, 9, 10, 11), "Morning",
                                             ifelse(peakerop$hour %in% c(12, 13, 14, 15), "Afternoon",
                                                    ifelse(peakerop$hour %in% c(16, 17, 18, 19), "Evening",
                                                            ifelse(peakerop$hour %in% c(20, 21, 22, 23), "Night",
                                                                    NA))))))

## test

#create a column where 1 means yes it is a weekday
peakerop$weekday_num <- ifelse(wday(peakerop$date, week_start = 1) <= 5, 1, 0)

write.csv(peakerop, "data/processed/peakerop_clean.csv", row.names = FALSE)

table(peakerop $operating)

##
##      0      1
## 11647 1505
## 1505/13152

## [1] 0.1144313

peakersummer <- peakerop |> dplyr::filter(season == "Summer")
table(peakersummer$operating)
```

```
##
##      0      1
## 3021 1395
1395/4416

## [1] 0.3158967
peakersummeron <- peakersummer |> dplyr::filter(operating == 1)
n_distinct(peakersummeron$date)

## [1] 80
n_distinct(peakersummer$date)

## [1] 184
80/184*100

## [1] 43.47826
```